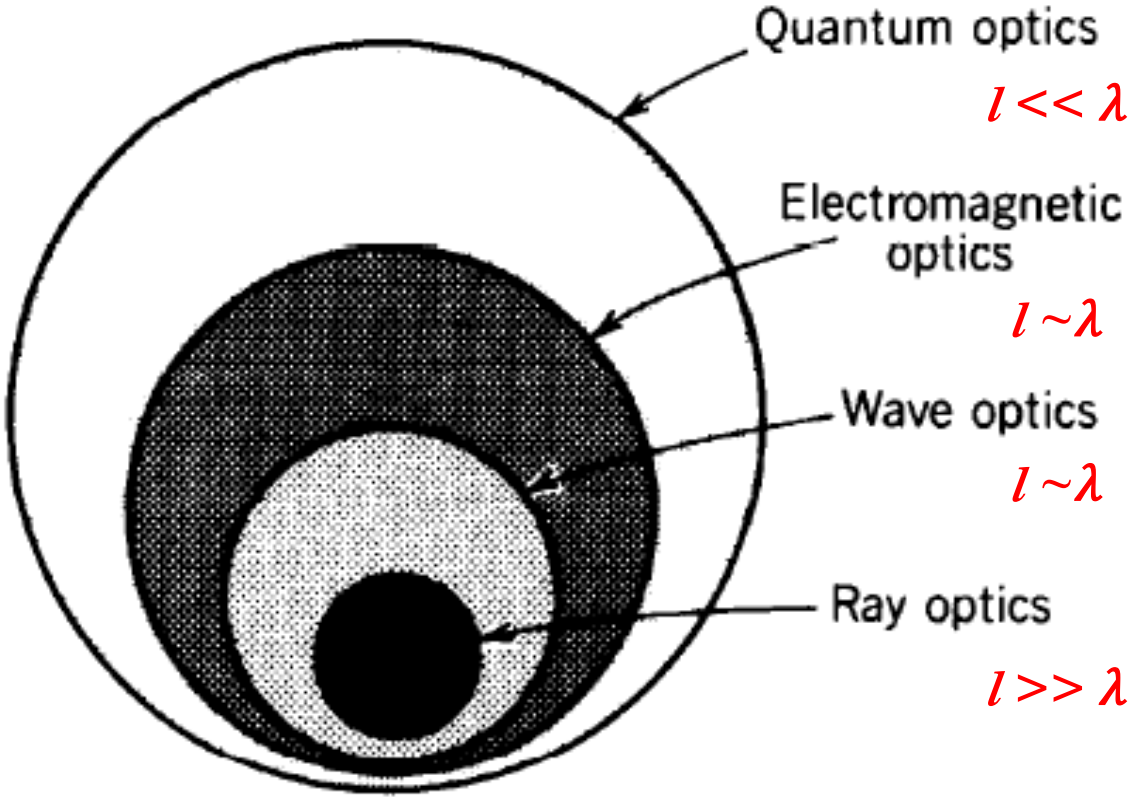
Online Office Hours

https://meet.goto.com/msunaidi/z_office-time

Problems Involving Light

1) Size-wavelength relationship.
2) Depth of analysis.



Use the "photon" concept
light-material interaction
on atomic scale

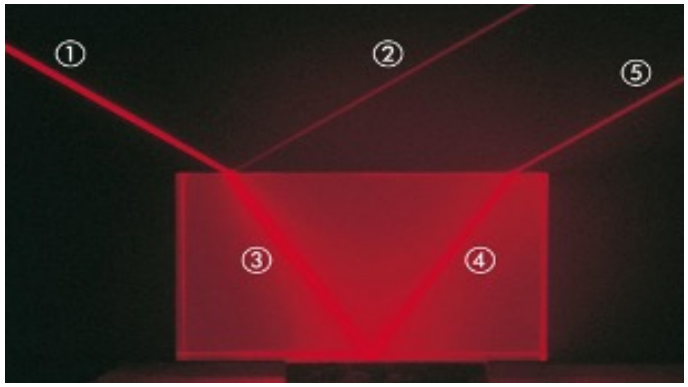
Use EM wave
representation
Almost all problems
Material interaction

Use basic wave
representation
More but limited analysis

Draw lines
Use geometry (angles)
Only basic analysis

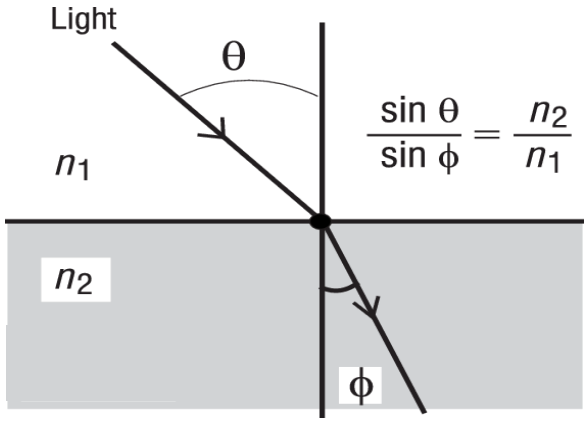
The Refractive Index

The refractive index n is the parameter that optically characterizes materials.









Claudius Ptolemy 150

Willebrord Snell 1620

Angle in air	Angle in water
10°	8°
20°	15-1/2°
30°	22-1/2°
40°	29°
50°	35°
60°	40-1/2°
70°	45-1/2°
80°	50°

Substance	Refractive index
Air	1.00029
Water	1.33
Ethyl alcohol	1.36
Fused quartz	1.46
Glycerine	1.47
Glass	1.45-1.70
Oil	1.50
Zircon	1.92
Diamond	2.42

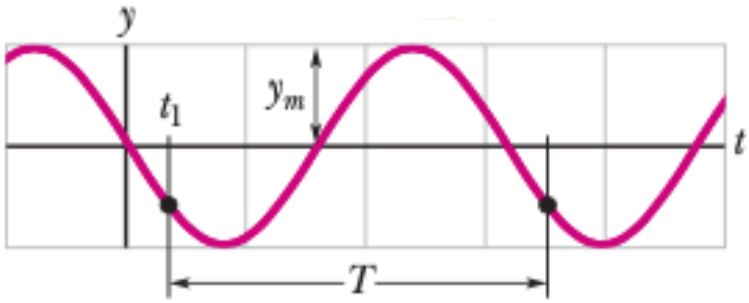
Ray and wave optics n is real.

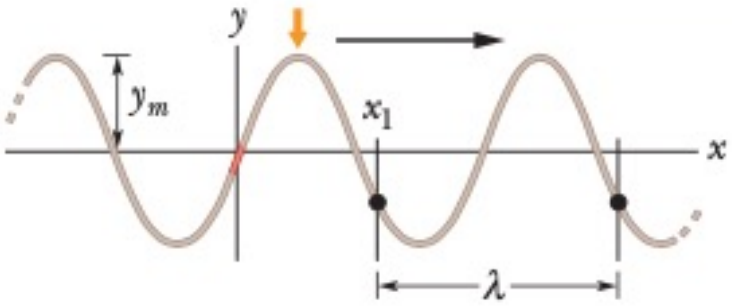
EM wave optics n is complex.

$$v = \frac{c}{n}$$

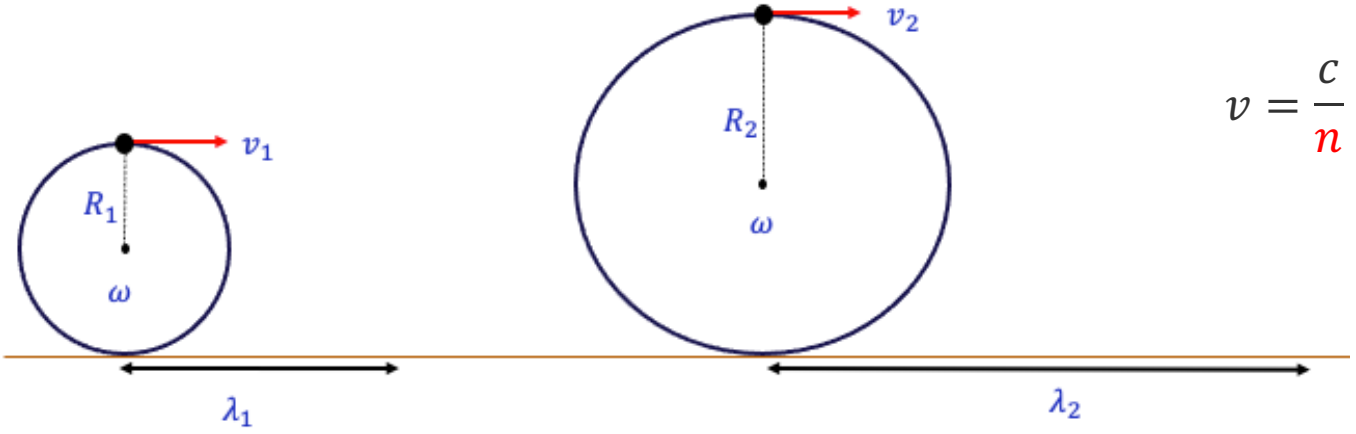
The Wavelength

The wavelength is defined as the distance the wave takes to make one spatial cycle.





Analogy (not drawn to scale)



$$v = \frac{c}{n}$$


Ray (Geometrical)
Optics

Foundation

Ray optics works well for problems involving light interaction with large objects and distances (the dimensions of the smallest feature of the object and the distances involved are much larger than the wavelength $l >> \lambda$).

In such situations, light makes traces depicted by rays (straight lines with arrows for direction). The analysis is based on solving the geometry generated by light path.

The guiding principles are:

1) Fermat's least-time principle, from which Snel's equations are derived.
2) Material is represented by the refractive index (velocity ratio).
3) No colors (wavelengths) are featured.
4) Amount of reflection/transmission is not featured.
5) Geometrical relations and techniques are applied.

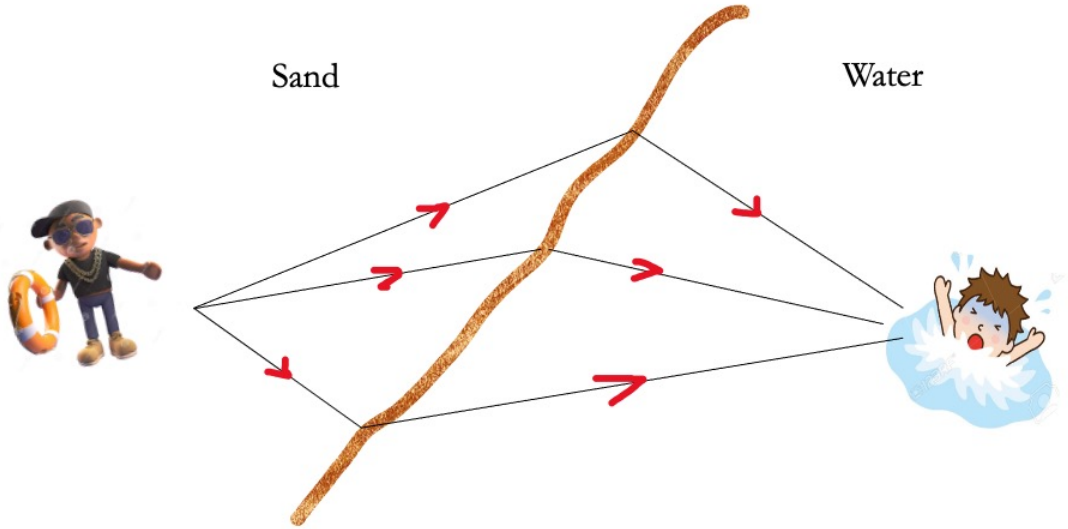
Snel's Laws of reflection and refraction.

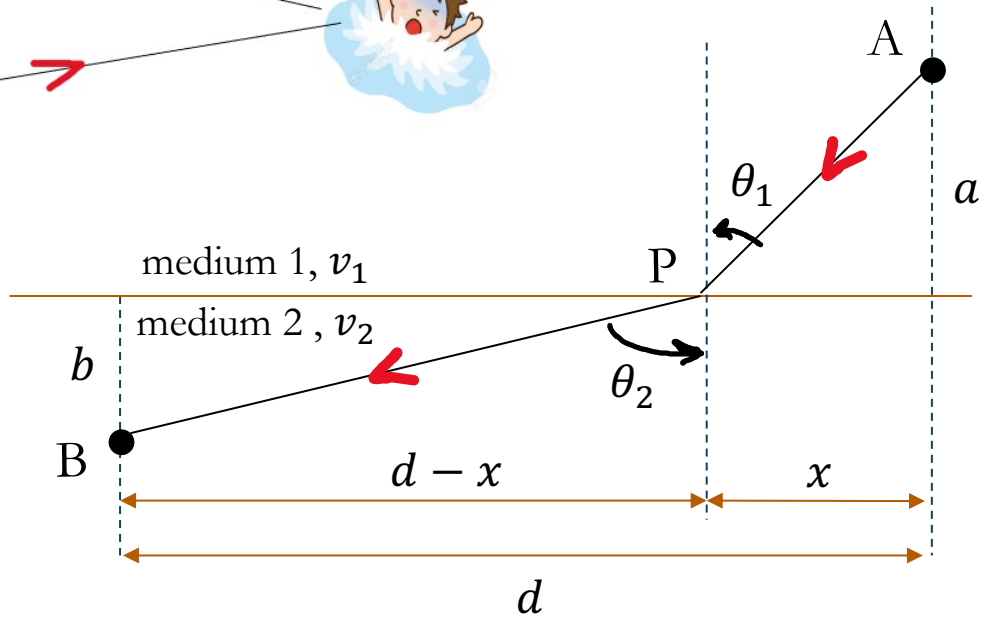
$$n_1 \sin\theta_i = n_2 \sin\theta_t$$

$$\sin\theta_i = \sin\theta_r$$

Fermat's Principle

Least time to reach target.





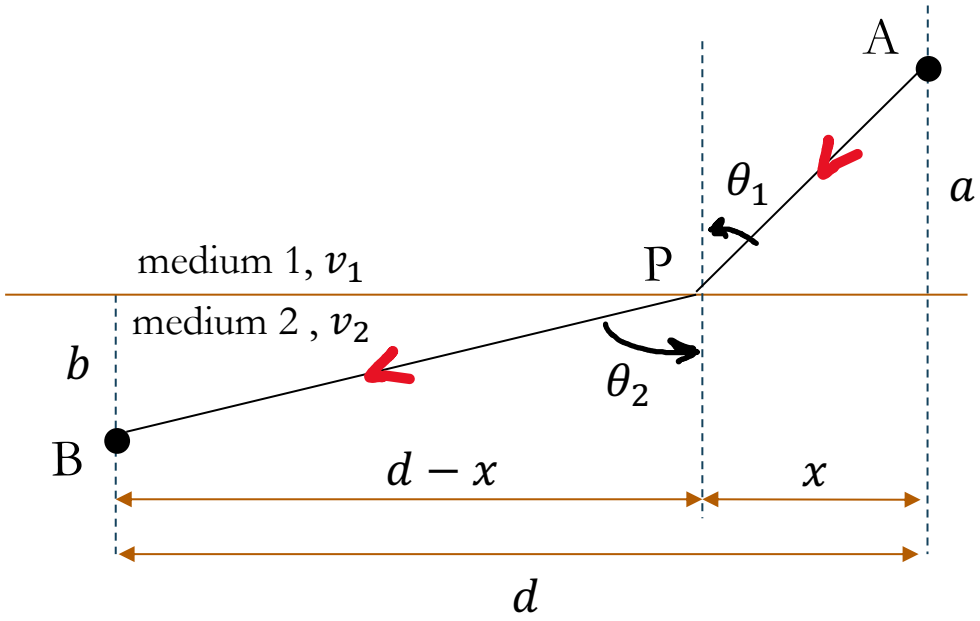
Fermat's Principle

Least time to reach target.

$$t_1 = \frac{AP}{v_1} = \frac{\sqrt{x^2 + a^2}}{v_1}$$

$$t_2 = \frac{PB}{v_2} = \frac{\sqrt{(d-x)^2 + b^2}}{v_2}$$

$$t = t_1 + t_2$$



$$t(x) = \frac{\sqrt{x^2 + a^2}}{v_1} + \frac{\sqrt{(d-x)^2 + b^2}}{v_2}$$

$$\frac{dt}{dx} = \frac{\sin\theta_1}{v_1} - \frac{\sin\theta_2}{v_2}$$

$$\frac{dt}{dx} = 0 \quad \Rightarrow \quad \sin\theta_2 = \sin\theta_1 \cdot \frac{v_2}{v_1} = \sin\theta_1 \cdot \frac{n_1}{n_2}$$

Snel's Law of Refraction $n_1 \sin\theta_1 = n_2 \sin\theta_2$

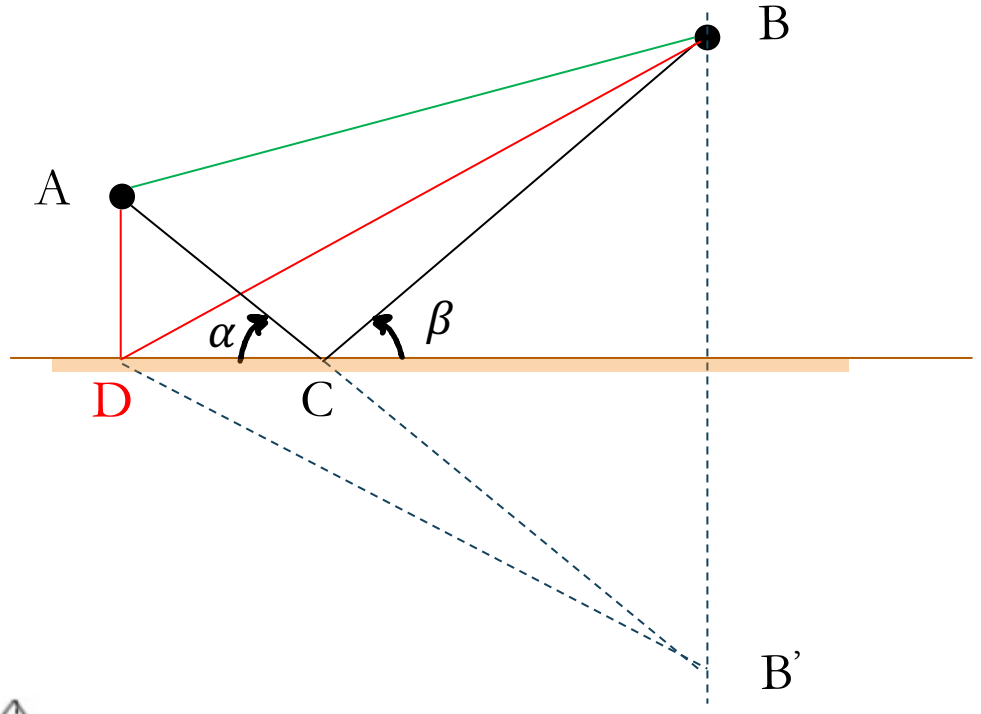
Snel's Law of Reflection $\sin\theta_i = \sin\theta_r$

Fermat's Principle

The mirror

For a homogeneous medium, least time is least distance.

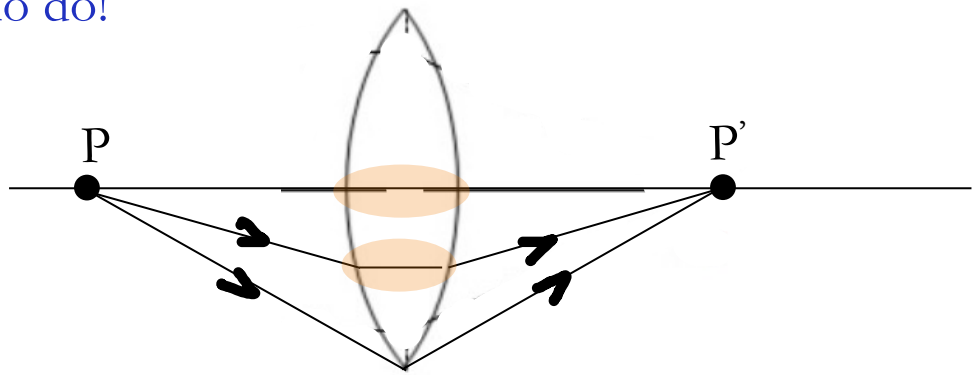
Least distance occurs at point C.



The lens

For focusing, make all possible light paths take equal time.

A square shape will no do!



The Prism

The properties of a glass flat prism can be determined from geometry. The prism angle is A.

$$\frac{\sin\alpha_i}{\sin\alpha_o} = n \qquad \frac{\sin\beta_i}{\sin\beta_o} = \frac{1}{n}$$

$$(180 - A) + \alpha_o + \beta_i = 180$$

$$\beta_i = A - \alpha_o = \beta_o - b$$

$$\alpha_i = a + \alpha_o$$

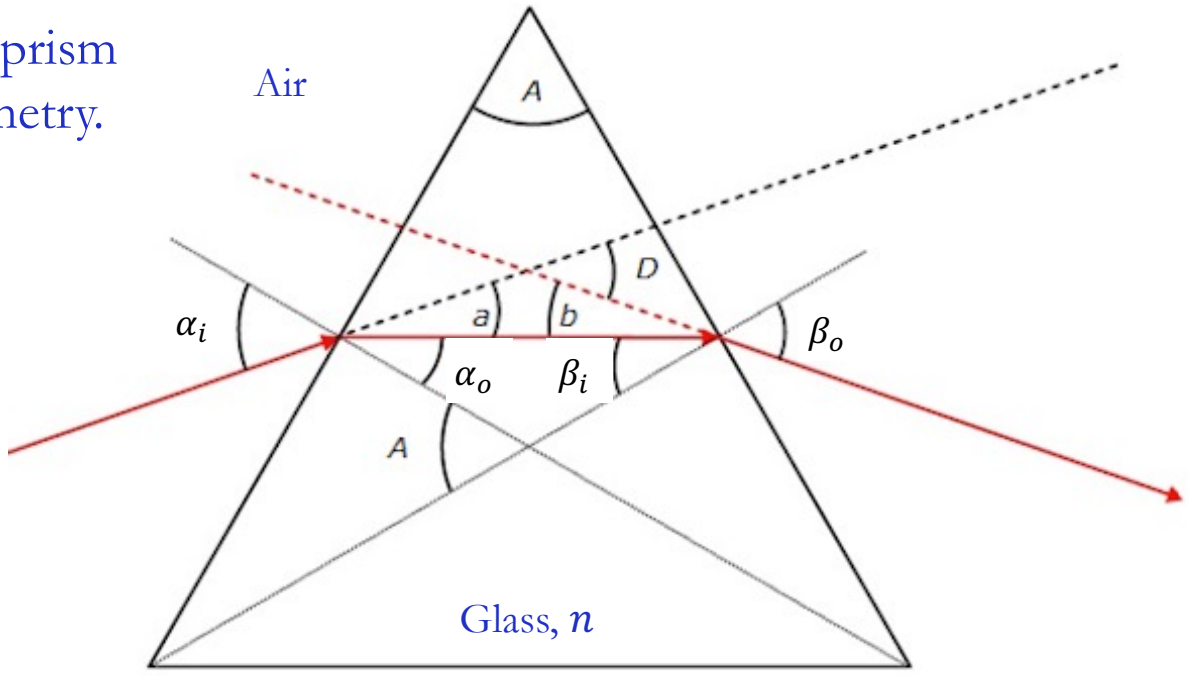
$$(180 - D) + a + b = 180$$

$$D = a + b = \alpha_i - \alpha_o + \beta_i - \beta_o$$

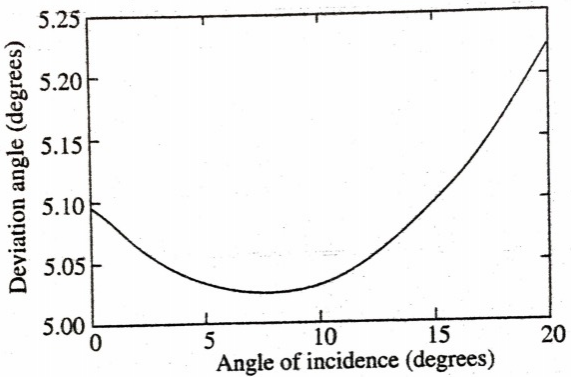
$$D = (\alpha_i + \beta_o) - A$$

An important parameter is the minimum deviation angle D_{min}.

$$\frac{n_{prism}}{n_o} = \frac{\sin[(A + D_{min})/2]}{\sin(A/2)}$$



Example $A = 10°, n_{prism} = 1.5, n_o = 1$

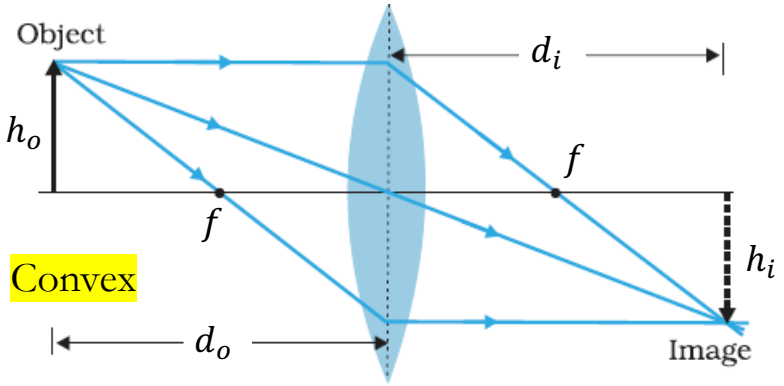


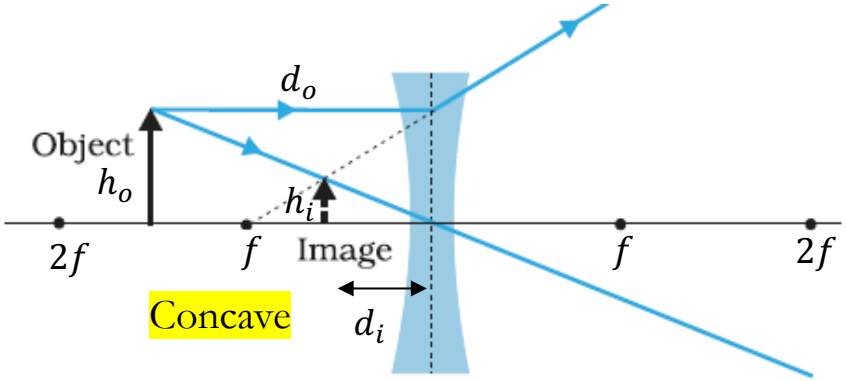
Lenses and Mirrors

Analysis of lenses is carried out based on the principle of refraction in geometrical optics, whereas mirrors are analyzed using reflection.

Lenses

The surfaces of lenses are usually spherical and designed in such away that a focal point is created. For this purpose, lenses are usually very thin.





Real images	Virtual images
Detectable on a screen	Cannot be detected on a screen
Formed by real rays	Formed by extensions.
Opposite side of object	Same side as object

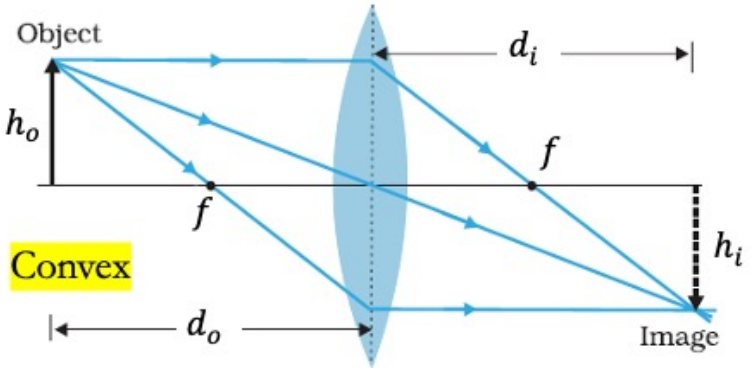
Lenses and Mirrors

The Lens Equation

A relation between the object and image locations with respect to the center line of the lens can be found geometrically. It works for both lens types.

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$$

$$f = R/2$$



Lens type	f	d_o	d_i	$M = h_i/h_o$	Image
Convex	+	$d_o > 2f$	$2f > d_i > f$	Reduced inverted	Real
		$2f > d_o > f$	$d_i > 2f$	Enlarged inverted	Real
		$f > d_o > 0$	$\|d_i\| > d_o$	Enlarged	Virtual

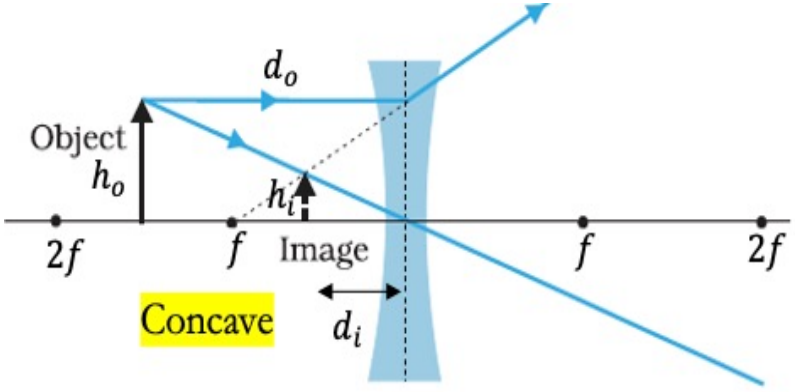
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		$2f > d_o > f$	$d_i > 2f$	Enlarged inverted	Real
		$f > d_o > 0$	$\lvert d_i \rvert > d_o$	Enlarged	Virtual
Concave	-	$d_o > 0$	$f > \lvert d_i \rvert > 0$	Reduced	Virtual

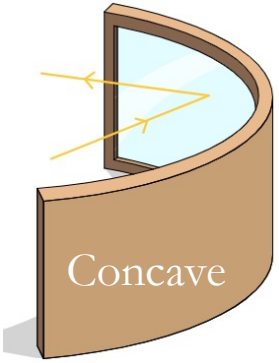
Lenses and Mirrors

The Lens Equation

A relation between the object and image locations with respect to the center line of the lens can be found geometrically. It works for both lens types.

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$$

$$f = R/2$$





Mirror type	Lens type	f	d_o	d_i	$M = h_i/h_o$	Image
Concave	Convex	+	$d_o > 2f$	$2f > d_i > f$	Reduced inverted	Real
			$2f > d_o > f$	$d_i > 2f$	Enlarged inverted	Real
			$f > d_o > 0$	$\|d_i\| > d_o$	Enlarged	Virtual
Convex	Concave	-	$d_o > 0$	$f > \|d_i\| > 0$	Reduced	Virtual

Lenses and Mirrors

The Lens Equation

A relation between the object and image locations with respect to the center line of the lens can be found geometrically. It works for both lens types.

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i} \qquad f = R/2$$

Example. A 3-cm object is placed 32 cm away from a convex lens with focus length of 8 cm. Describe the image.

$$\frac{1}{8} = \frac{1}{32} + \frac{1}{d_i} \qquad d_i = 11\ cm,$$

$$h_i = -1\ cm, \qquad M = \frac{1}{3}$$

Lens type	f	d_o	d_i	M	Image
Convex	+	$d_o > 2f$	$2f > d_i > f$	Reduced inverted	Real
		$2f > d_o > f$	$d_i > 2f$	Enlarged inverted	Real
		$f > d_o > 0$	$\lvert d_i \rvert > d_o$	Enlarged	Virtual

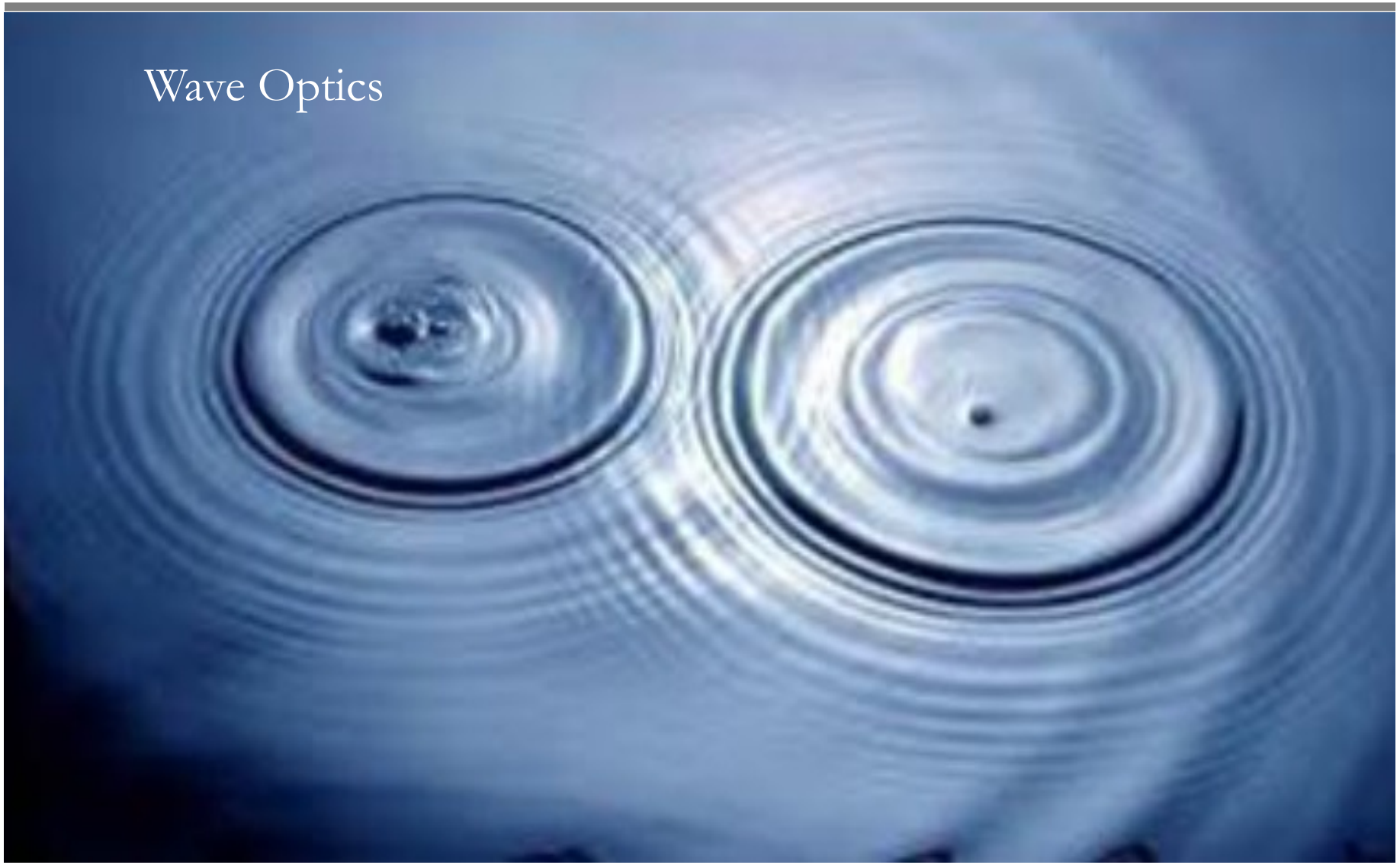
Mirrors

Mirrors are single-sided lenses (other surface is metal-coated). The same lens equation is applicable for mirrors.

Example. A store watch mirror with focus length of 3 m. The store is 7 m deep.

$$\frac{-1}{3} = \frac{1}{7} + \frac{1}{d_i} \qquad d_i = -2.1\ m, \quad M = 0.3$$

Convex		-	$d_o > 0$	$f > \lvert d_i \rvert > 0$	Reduced	Virtual


Wave Optics

Foundation

Wave optics can better describe the different phenomena involving light, mainly those associated with the wave nature of light such as interference. Objects comparable in size to optical wavelength can be treated more adequately than ray optics. However, the description of light-matter interaction is limited.

In wave optics, light is considered as a wave traveling in space with a defined amplitude, speed and wavelength, just like mechanical (sound) waves. Wave polarization is not relevant.

The guiding principles are:

1) Light is described by the scalar wave equation.
2) Material is represented by the refractive index (velocity ratio) and attenuation.
3) The light wave has a specified amplitude, velocity and wavelength.
4) Amount of reflection/transmission and wave polarization cannot be determined.

The scalar wave equation

$$\nabla^2 u - \frac{1}{v^2}\frac{d^2 u}{dt^2} = 0$$

The Scalar Wave Equation

The scalar wave equation

$$\nabla^2 U - \frac{1}{v^2}\frac{d^2U}{dt^2} = 0$$

$U(r,t)$ is wave function

Assume that the wave function is time-harmonic

$$U(r,t) = U(r)\, e^{j\omega t}$$

The scalar wave equation becomes

$$\nabla^2 U(r) + \left(\frac{\omega}{v}\right)^2 U(r) = 0$$

$$(\nabla^2 + k^2)\, U(r) = 0$$

Put $k = \frac{\omega}{v}$ Wave number (rad/m)

$$k = \frac{\omega}{v} = \frac{2\pi f}{v} = \frac{2\pi}{\lambda}$$

$$v = \frac{v_o}{n}$$

$$k = nk_o$$

$$\lambda = \frac{\lambda_o}{n}$$

A general solution is of the form

$$U(r) = a(r) \cdot e^{j\phi(r)}$$

A general time-harmonic solution is of the form

$$U(r,t) = a(r) \cdot e^{j\phi(r)} \cdot e^{j\omega t}$$

An instantaneous solution is of the form

$$u(r,t) = a(r) \cdot \cos[\omega t + \phi(r)]$$

$$\phi(r) = kr$$

$$u(r,t) = \frac{1}{2}[U(r,t) + U^*(r,t)]$$

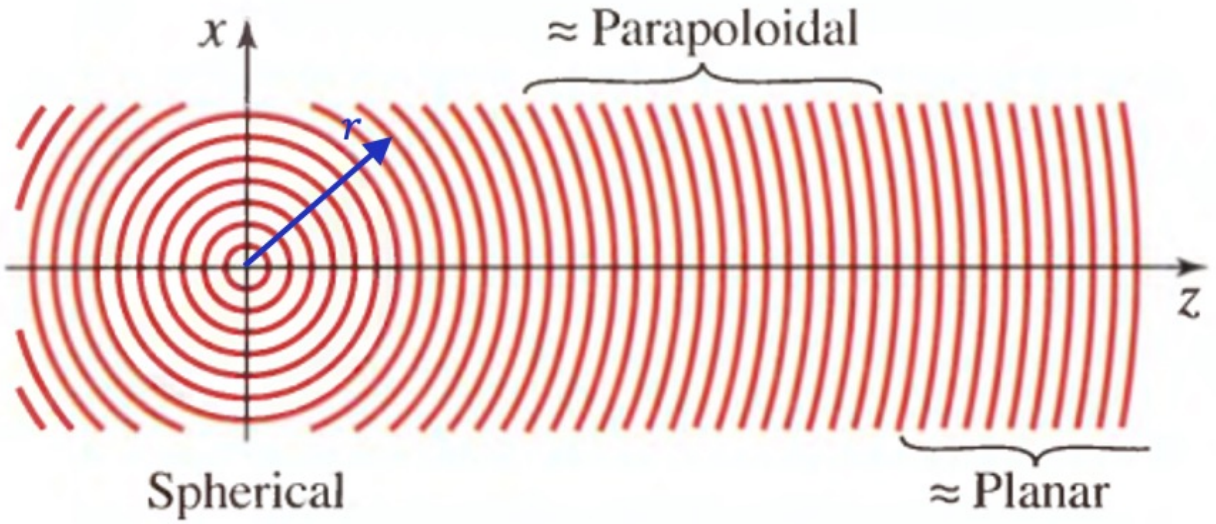
Some Wave Types

A number of possible forms of the solutions $U(r) = a(r) \cdot e^{j\phi(r)}$

Spherical waves $U(r) = \frac{A}{r} e^{-jkr}$

Plane waves $U(r) = A\, e^{-jkz}$

Parapoloidal waves $U(r) = \frac{A}{z} e^{-jkz}\, e^{-jk\left(\frac{x^2+y^2}{2z}\right)}$



Wave Intensity and Power

1) Optical intensity of a wave is given by

$$I(r) = |U(r)|^2$$

Time-harmonic

Monochromatic light

Such that

$$U(r) = \sqrt{I(r)} \cdot e^{j\phi(r)}$$

Instantaneous intensity

$$I(r,t) = < U(r,t) \cdot U^*(r,t) >$$

2) Wave power is given by

$$P = \int_A I(r)\, dA$$

Spherical waves

$$U(r) = \frac{\sqrt{I}}{r} e^{-jkr}$$

Plane waves

$$U(r) = \sqrt{I}\, e^{-jkz}$$

Interference (1)

1) Two monochromatic plane light waves, same wavelength.

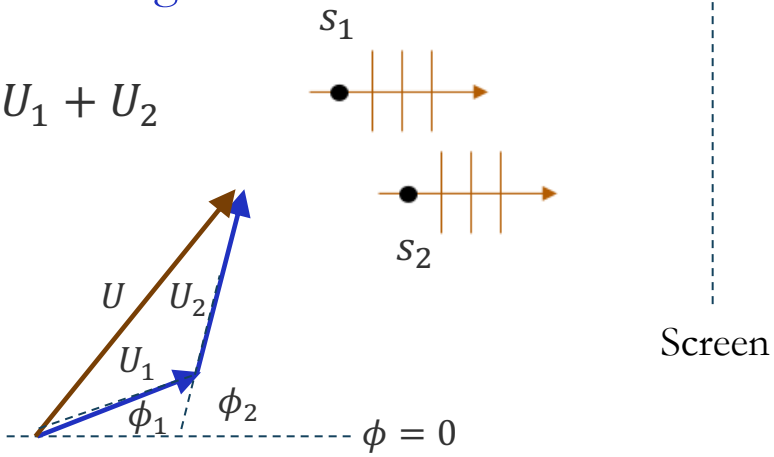
$$U_1 = \sqrt{I_1} \cdot e^{j\phi_1} \qquad U_2 = \sqrt{I_2} \cdot e^{j\phi_2} \qquad U(r) = U_1 + U_2$$

$$I(r) = |U(r)|^2 = |U_1 + U_2|^2$$

$$I(r) = |U_1|^2 + |U_2|^2 + U_1 U_2^* + U_1^* U_2$$

$$I(r) = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos(\phi_2 - \phi_1)$$

Interference equation



Interference (1)

1) Two monochromatic plane light waves, same wavelength.

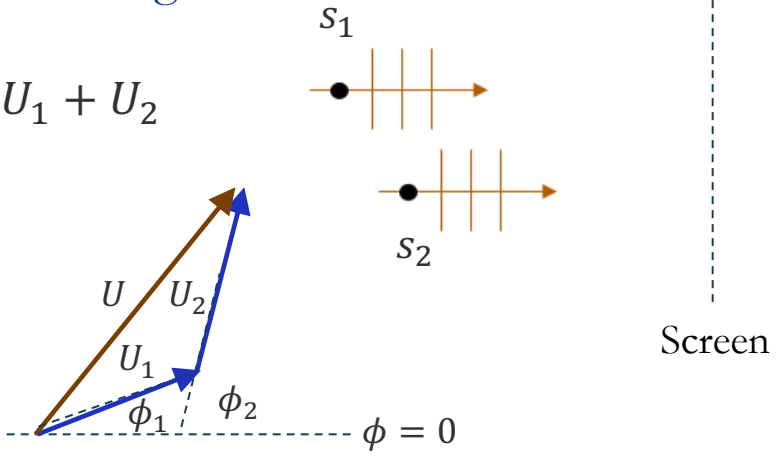
$$U_1 = \sqrt{I_1}\cdot e^{j\phi_1} \qquad U_2 = \sqrt{I_2}\cdot e^{j\phi_2} \qquad U(r) = U_1 + U_2$$

$$I(r) = |U(r)|^2 = |U_1 + U_2|^2$$

$$I(r) = |U_1|^2 + |U_2|^2 + U_1U_2^* + U_1^*U_2$$

$$I(r) = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos(\phi_2 - \phi_1)$$

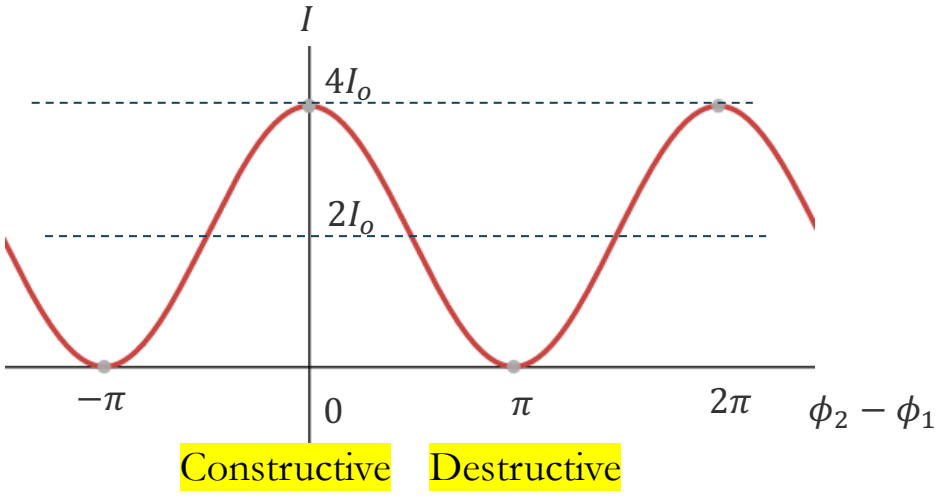
Interference equation



Example. Draw the interference pattern for two sources with $I_1 = I_2 = I_o$

$$I = 2I_o[1 + \cos(\phi_2 - \phi_1)]$$

What is the pattern on the screen?



Interference (1)

Where does the phase come from?

1) Sources are out-of-phase.

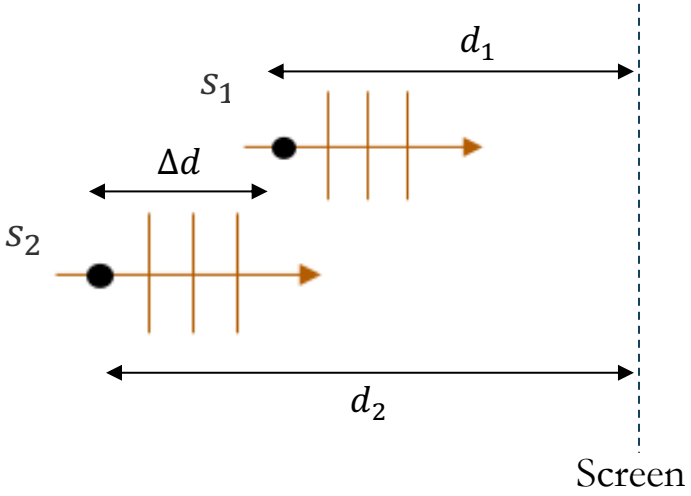
2) Distance.

$$U_1 = \sqrt{I_o} \cdot e^{-jkd_1} \qquad U_2 = \sqrt{I_o} \cdot e^{-jkd_2}$$

$$I = 2I_o[1 + \cos(k\Delta d)]$$

$$k\Delta d = \frac{2\pi}{\lambda}\Delta d \qquad \Delta d = \pm m\lambda \quad \text{Constructive}$$

$$\Delta d = \pm\frac{(2m+1)\lambda}{2} \quad \text{Destructive} \qquad m = 0, 1, 2, \ldots$$



3) Properties of path (n).

$$U_1 = \sqrt{I_o} \cdot e^{-jk_1 d} \qquad U_2 = \sqrt{I_o} \cdot e^{-jk_2 d}$$

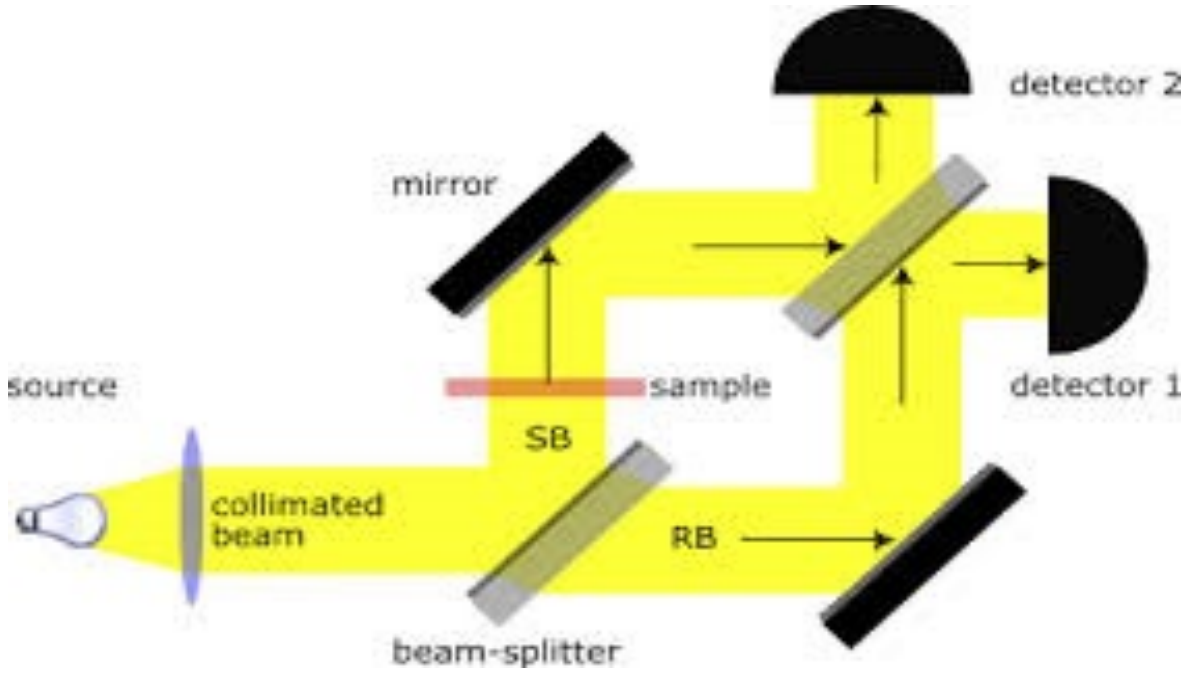
$$I = 2I_o[1 + \cos(k_2 - k_1)d]$$

$$(k_2 - k_1)d = \frac{2\pi}{\lambda} d(n_2 - n_1) = \frac{2\pi d}{\lambda}\Delta n$$

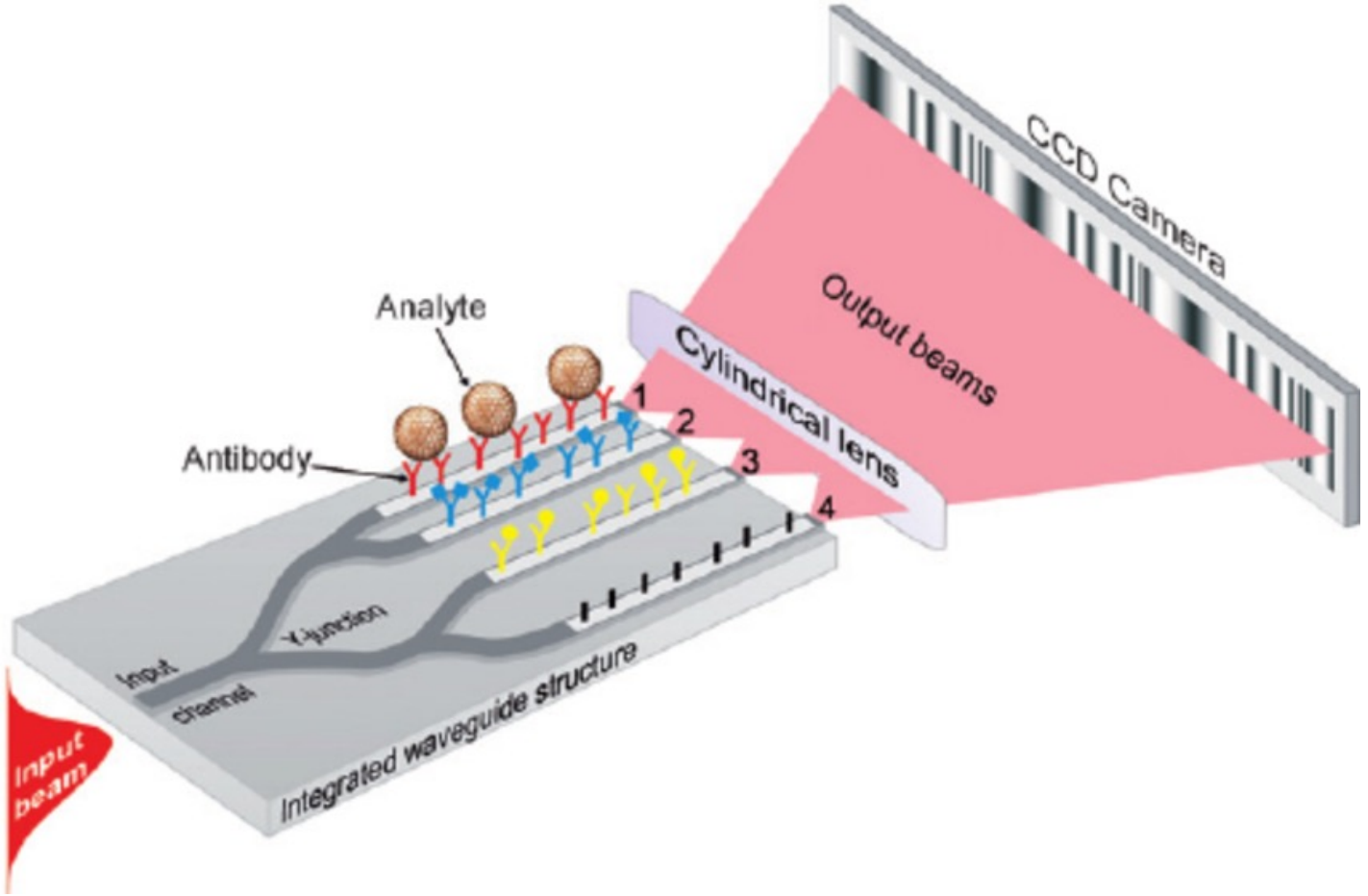
Applications: Interferometry

1) Strain/stress
2) Index change (uniformity)
3) Layer thickness.

Interferometer



Interferometry in Biosensing



Interference (2)

2) Two monochromatic plane light waves, same wavelength, at oblique angle.

$$U_1 = \sqrt{I_o} \cdot e^{-jkz} \qquad U_2 = \sqrt{I_o} \cdot e^{-jk(x \sin\theta + z\cos\theta)}$$

At the screen,

$$U_1 = \sqrt{I_o} \qquad U_2 = \sqrt{I_o} \cdot e^{-jkx \sin\theta}$$

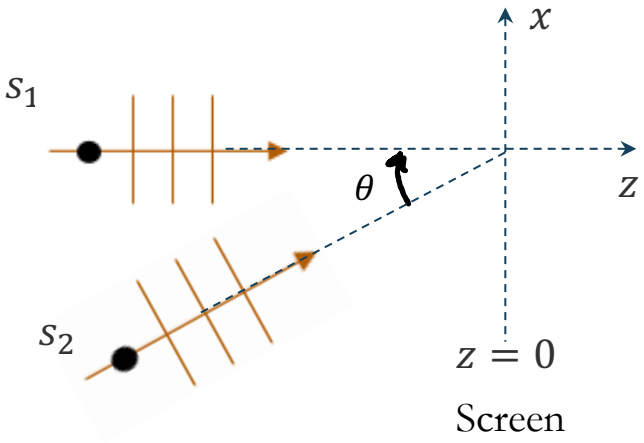
$$I = 2I_o[1 + \cos(kx \sin\theta)]$$

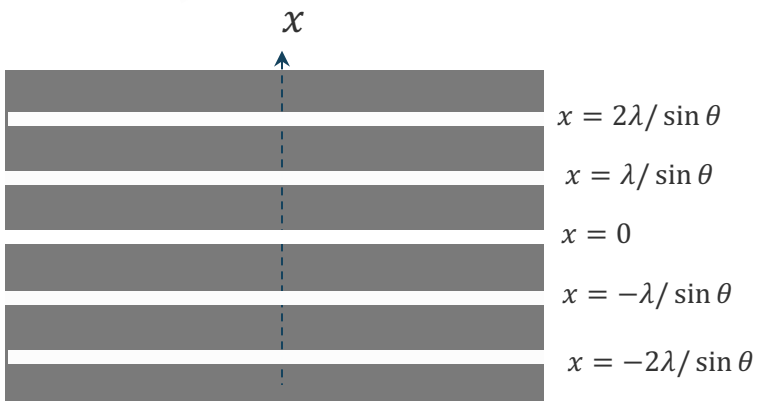
$$kx \sin\theta = m(2\pi) \quad \text{Maxima} \quad m = 0, 1, 2, \ldots$$

$$x_{max} = \frac{m(2\pi)}{k \sin\theta} = m\frac{\lambda}{\sin\theta}$$

$$kx \sin\theta = (2m+1)\pi \quad \text{Minima}$$

$$x_{min} = \frac{(2m+1)\pi}{k \sin\theta} = \left(\frac{2m+1}{2}\right)\frac{\lambda}{\sin\theta}$$





Applications: Lithography

Example. Calculate the required angle to produce a 1 μm feature using blue light ($\lambda = 450\ nm$).

Q & A

Send me an email if you have any questions.